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Using the Slope Formula

This formula from the GED math formula sheet uses subtraction to find slope.

slope (m) = $\frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are two points on the coordinate plane

Example

Hide

What is the slope of the line that passes through the points (1, 1) and (−2, 3)?

Step 1

Decide which point will be (x_1, y_1) .

Let (1, 1) be (x_1, y_1) and (−2, 3) be (x_2, y_2) .

Step 2

Substitute these values into the formula and simplify.

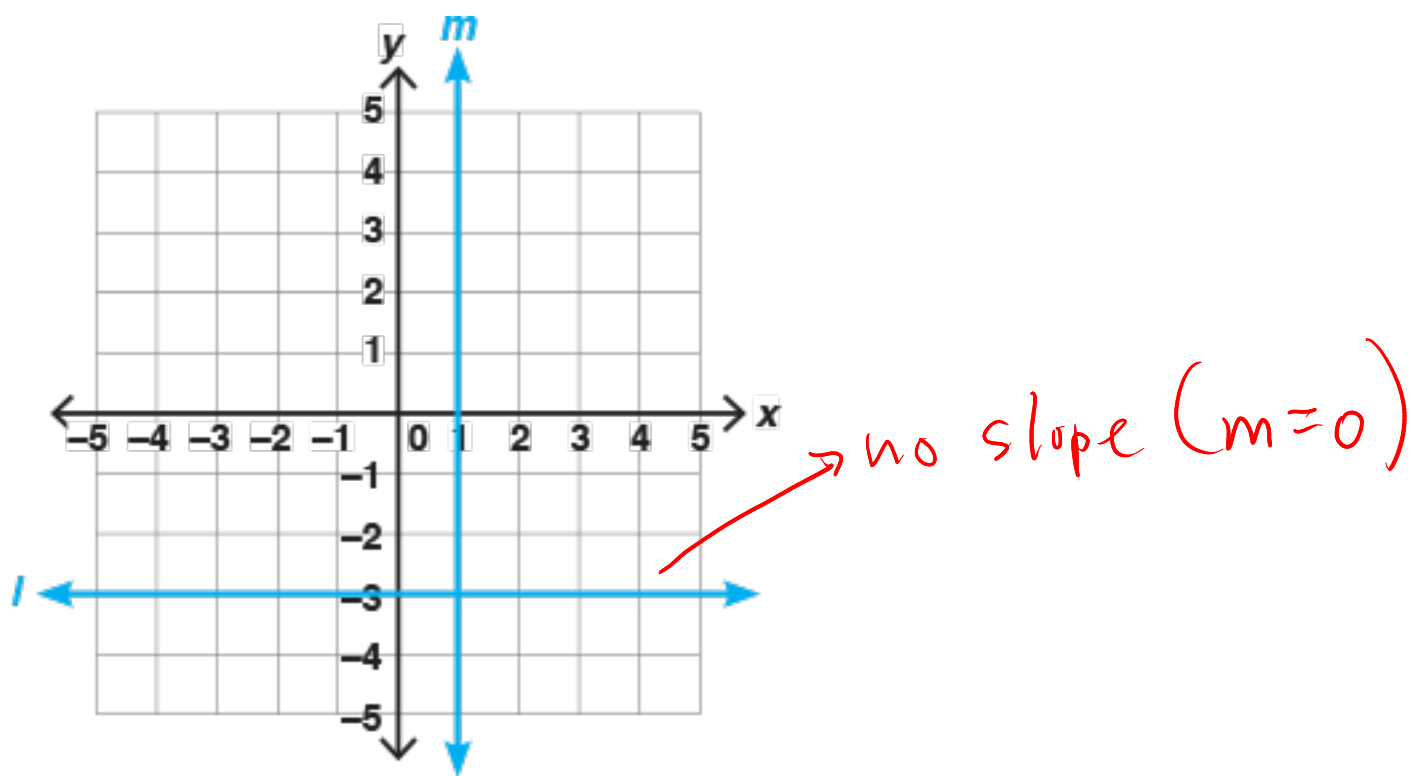
$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{-2 - 1} = \frac{2}{-3} = -\frac{2}{3}$$

The slope is $-\frac{2}{3}$.

or, if (x_1, y_1) is $(-2, 3)$:

$$m = \frac{1 - 3}{1 - (-2)} = \frac{-2}{3} \checkmark$$

There are two kinds of lines with special slopes.



Look at line l . It doesn’t rise at all. For every unit of run, the line rises 0 units. The slope is $\frac{0}{1}$, or 0.

Look at line m . It only rises; it never runs. For every unit of rise, the line runs 0 units. You could write the slope as $\frac{1}{0}$, but division by 0 has no meaning in math. The slope of a vertical line is undefined.

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Lesson 2: Slope

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